

GALVESTON, TX, USA

WMO: 722420

Lat: 29.273N

Lon: 94.859W

Elev: 2

StdP: 101.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12923

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.5	(c) 4.2	(d) -5.0	(e) 2.5	(f) 5.7	(g) -2.5	(h) 3.1	(i) 8.2	(j) 12.9	(k) 8.8	(l) 12.0	(m) 8.4	(n) 6.9	(o) 20	(p) 0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.3	(c) 33.4	(d) 26.3	(e) 32.7	(f) 26.4	(g) 32.3	(h) 26.4	(i) 27.9	(j) 31.2	(k) 27.5	(l) 30.8	(m) 27.2	(n) 30.5	(o) 5.2	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	22.9	29.3	26.5	22.1	29.3	26.2	21.7	29.3	88.5	31.4	86.8	31.0	85.5	30.5	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.5	(b) 10.3	(c) 9.0	(e) -0.2	(f) 35.6	(g) 2.1	(h) 1.7	(i) -1.7	(j) 36.8	(k) -3.0	(l) 37.8	(m) -4.1	(n) 38.7	(o) -5.7	(p) 39.9		
(4) 11.5	(5) 10.3	(6) 9.0	(7) -0.2	(8) 35.6	(9) 2.1	(10) 1.7	(11) -1.7	(12) 36.8	(13) -3.0	(14) 37.8	(15) -4.1	(16) 38.7	(17) -5.7	(18) 39.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	13.0	14.8	18.1	21.6	25.4	28.5	29.3	29.7	27.8	23.9	18.5	14.3
	DBStd	6.78	4.59	4.38	3.69	2.88	2.11	1.47	1.58	1.40	2.06	3.58	4.38	4.57
	HDD10.0	53	24	11	2	0	0	0	0	0	0	0	2	13
	HDD18.3	532	169	110	48	8	0	0	1	0	0	5	52	138
	CDD10.0	4474	119	146	254	348	476	554	597	610	535	431	257	147
	CDD18.3	1910	5	11	42	107	218	304	339	352	285	177	56	14
	CDH23.3	19317	2	6	48	344	1692	3583	4346	4557	3179	1393	155	12
CDH26.7	7197	0	0	2	24	304	1351	1921	2142	1143	302	7	0	
Wind	WSAvg	5.0	5.2	5.3	5.4	5.7	5.6	5.0	4.4	4.3	4.4	4.9	5.0	5.1
Precipitation	PrecAvg	1102	87	58	66	61	81	107	89	117	155	93	86	94
	PrecMax	1632	273	212	272	264	280	375	444	677	618	300	363	281
	PrecMin	505	5	0	2	0	0	5	0	3	7	1	10	13
	PrecStd	286	62	46	59	55	68	90	78	113	111	75	67	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.3	24.3	26.1	28.6	31.1	33.7	33.6	34.7	34.0	31.4	27.6	24.8
		MCWB	19.4	20.5	21.3	22.1	24.7	25.6	26.2	26.2	25.6	25.8	23.6	21.6
	2%	DB	21.7	22.7	24.6	27.3	29.9	32.4	32.9	33.6	32.4	30.3	26.3	23.0
		MCWB	18.8	20.2	21.3	22.6	24.7	26.0	26.5	26.5	26.0	25.5	22.8	20.9
	5%	DB	20.3	21.4	23.6	26.2	29.0	31.7	32.4	32.9	31.7	29.2	25.1	22.1
		MCWB	18.3	19.2	20.7	22.4	24.5	26.0	26.5	26.6	26.0	24.8	22.3	20.4
	10%	DB	19.0	20.2	22.6	25.3	28.3	31.1	32.0	32.4	31.0	28.3	24.0	21.0
		MCWB	17.4	18.5	20.1	21.9	24.2	25.9	26.5	26.5	25.6	24.3	21.5	19.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	22.0	23.0	24.9	26.7	28.0	28.0	28.4	28.0	27.5	25.1	22.8
		MCDB	22.1	23.5	24.7	26.6	29.0	31.1	31.7	32.0	31.1	29.3	26.6	24.0
	2%	WB	19.7	21.1	22.2	24.1	26.0	27.6	27.6	27.9	27.5	26.7	23.9	21.8
		MCDB	20.8	22.2	23.7	26.0	28.3	30.4	31.2	31.6	30.0	28.8	25.3	22.7
	5%	WB	18.9	19.9	21.5	23.5	25.5	27.1	27.3	27.4	27.0	26.0	23.0	21.0
		MCDB	19.8	20.8	22.9	25.4	28.0	29.9	30.9	31.2	29.7	28.3	24.8	21.7
	10%	WB	17.9	18.9	20.9	22.9	25.1	26.7	26.9	27.1	26.6	25.3	22.2	19.6
		MCDB	18.5	19.8	22.2	24.7	27.6	29.7	30.5	30.9	29.4	27.9	23.7	20.5
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.2	5.9	5.7	5.1	4.9	5.1	5.3	5.4	5.9	6.4	6.6
		MCDBR	7.2	6.6	5.6	5.6	5.3	5.4	5.5	5.8	5.7	5.5	6.1	7.1
		MCWBR	5.7	5.0	3.8	3.3	2.6	2.5	2.4	2.4	2.6	3.3	4.1	5.8
	5% WB	MCDBR	6.7	6.0	5.0	4.8	4.5	4.7	4.9	5.3	5.0	5.3	5.8	6.8
		MCWBR	6.0	5.1	3.8	3.5	2.6	2.6	2.5	2.6	2.6	3.9	4.5	6.2
Clear-Sky Solar Irradiance	taub	0.351	0.366	0.380	0.434	0.475	0.465	0.487	0.472	0.450	0.387	0.375	0.359	
	taud	2.510	2.479	2.426	2.276	2.200	2.285	2.244	2.287	2.312	2.476	2.456	2.494	
	Ebn at Noon	886	902	908	865	827	830	810	821	828	867	849	857	
	Edh at Noon	90	101	113	136	147	134	140	133	125	100	95	88	
All-Sky Solar Radiation	RadAvg	2.91	3.32	4.57	5.46	6.18	6.40	6.07	5.81	5.01	4.44	3.37	2.60	
	RadStd	0.29	0.54	0.39	0.41	0.40	0.46	0.56	0.45	0.53	0.52	0.34	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118	+97

Nomenclature: See separate page